

EE596H (Optimization Techniques): Half Semester Examination

Total Questions: 4, Total Marks: 45, Time: 180 Mins

1. For x, y both positive scalars, show that

$$ye^{x/y} = \max_{a>0} (a(x+y) - ya \cdot \ln a).$$

Use the above result to prove that the function f defined as

$$f(x, y) = \begin{cases} ye^{x/y} & \text{if } x > 0, y > 0, \\ +\infty & \text{otherwise,} \end{cases}$$

is convex.

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2. Consider the following problem:

$$\begin{aligned} &\text{minimize} && -3x_1^2 + x_2^2 + 2x_3^2 + 2(x_1 + x_2 + x_3) \\ &\text{subject to} && x_1^2 + x_2^2 + x_3^2 = 1, \end{aligned}$$

- (a) Is the above objective function a convex function?
 (b) Is the feasible set a convex set?
 (c) Is there any hidden convexity? Explain and elaborate by variable transformation if requires.
 (d) Derive the dual problem. Does strong duality hold? Explain and elaborate. If you are not able to solve fully, leave it at the point where you get stuck. However, give proper justification to your answer.

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Total: 15

3. Consider the problem

$$\text{minimize } f(\mathbf{x}) = 10x_1 + 3x_2$$

subject to

$$5x_1 + x_2 \geq 4, \quad x_1, x_2 = 0 \text{ or } 1.$$

- (a) Derive and sketch the dual function.
 (b) Is there any duality gap? Justify your answer.

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Total: 10

4. Consider the problem

$$\begin{aligned} &\text{minimize} && x_1^2 + 2x_2^2 + 2x_1x_2 + x_1 - x_2 - x_3 \\ &\text{subject to} && x_1 + x_2 + x_3 \leq 1, \\ &&& x_3 \leq 1. \end{aligned}$$

- (a) Is the problem convex?
 (b) Find a dual problem and solve it.
 (c) Using the dual optimal solution, find an optimal solution of the primal problem.

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Total: 15

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